

Floating Point Representation IEEE standard:

Single Precision $\rightarrow 32$ bits.

double Precision $\rightarrow 64$ bits.

For single precision:

sign bit $\rightarrow 1$ bit.

Exponent $\rightarrow 8$ bits.

Exponent
Mantissa $\rightarrow 23$ bits.

Bias for single Precision $\rightarrow 127$.

for double Precision:

sign bit $\rightarrow$ 1 bit

Exponent $\rightarrow$ 11 bits

Mantissa $\rightarrow$ 52 bits

Bias for double precision $\rightarrow 1023$.

Ex - Represent $(5.5)_{10}$ in single precision IEEE standard format.

5°5

5's
Binary equivalent of $(5's)_{10} = (101)_2$

Now, $101^{\circ}1 = 1^{\circ}011 \times 2^2 \rightarrow \text{exponent}$
 $\rightarrow \text{mantissa}$

→ mantissa

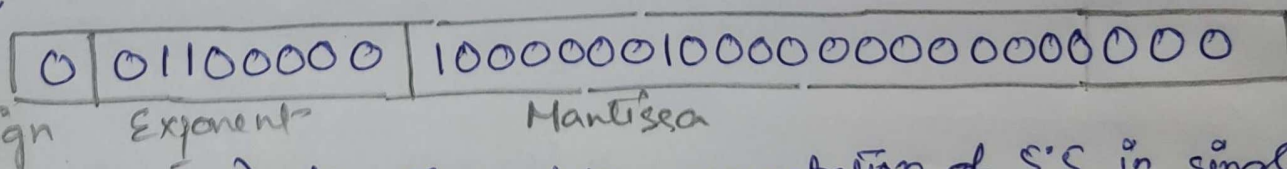
Here, $\text{sign} = 0$ [as it is a +ve number]

Mantissa = 011

Exponent = $2 + \text{bias} = 2 + 127 = 129$.

$$= 100000001 \text{ [Binary equivalent of 129]}$$

Thus,



is the required floating-point representation of S's in single precision IEEE standard form